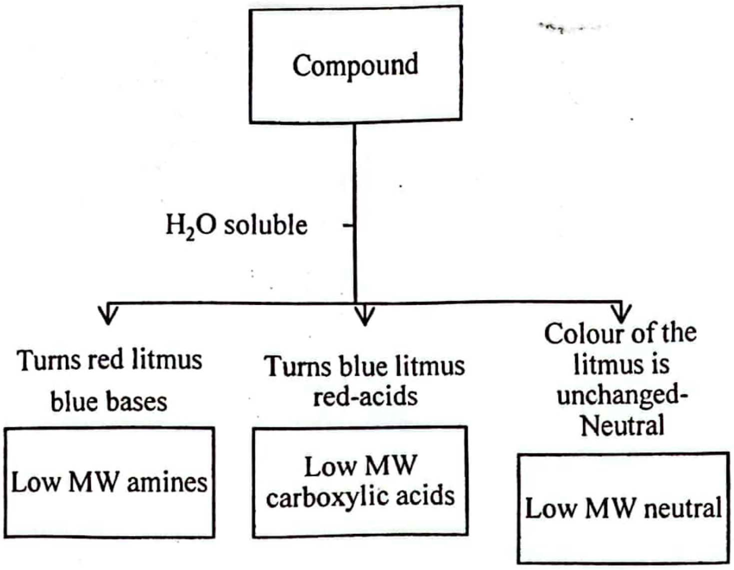
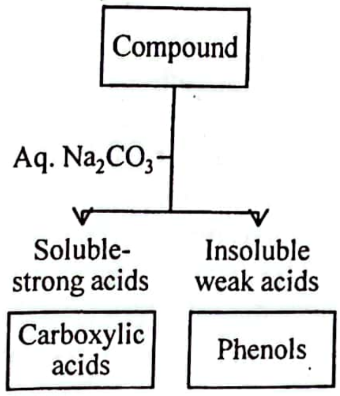
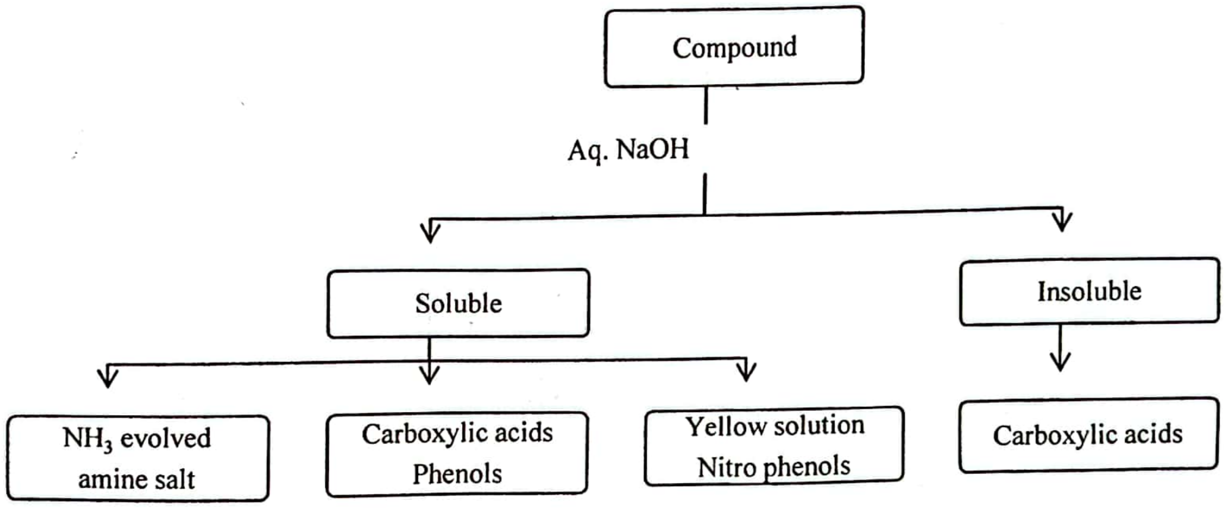
2. SOLUBILITY BEHAVIOUR.

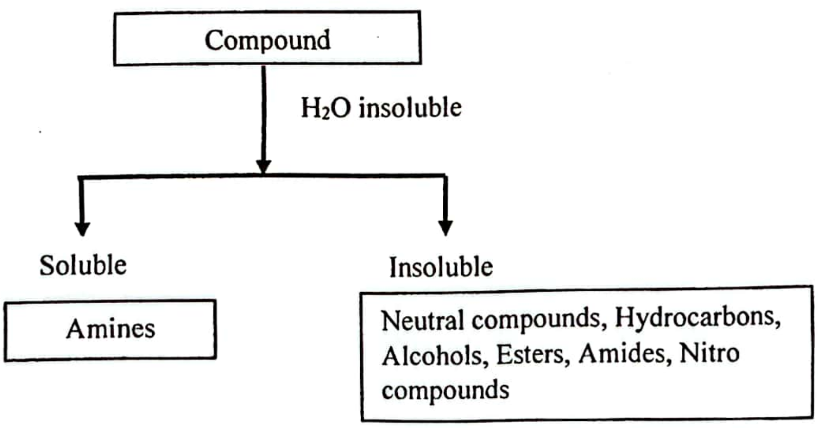
The solubility of an organic compound provides useful information regarding its structure and properties. In general, acidic compound s are soluble in bases and basic compounds are soluble in acids. Neutral compounds which are insoluble in water are also insoluble in both acids and bases.

Following charts indicates solvents in which compounds containing various functional groups are likely to dissolve.


Compound
H_2O soluble
Turns red litmus blue bases
Turns blue litmus red-acids
Colour of the litmus is unchanged-Neutral
Low MW amines
Low MW carboxylic acids
Low MW neutral


Compound
Aq. Na_2CO_3
Soluble-strong acids
Insoluble weak acids
Carboxylic acids
Phenols


Compound
Aq. NaOH
Soluble
Insoluble
NH_3 evolved amine salt
Carboxylic acids Phenols
Yellow solution Nitro phenols
Carboxylic acids


Compound
H_2O insoluble
Soluble
Insoluble
Amines
Neutral compounds, Hydrocarbons, Alcohols, Esters, Amides, Nitro compounds

Determine the solubility of an unknown compound in a particular solvent, by using only a small amount of the sample

Procedure: Place about 1 ml of the solvent in a small test tube and few crystals of the unknown solid or one drop of the unknown liquid directly into the solvent. Gently tap the test tube to ensure mixing and observe the disappearance of the liquid or solid, which indicate that solution is taking place. (Do not use a large amount of the sample. A common mistake in determining the solubility is testing with a quantity of the unknown too large to dissolve in the amount of solvent taken. Use small amount)

Low molecular weight compounds that contain up to 4 carbons and also contain O, N or S are often soluble in water. Almost any functional group containing these elements will lead to water solubility for low-molecular weight (C_4) compounds. When the ratio of the O, N or S atoms (polar functional groups) in a compound to the carbon atoms is increased, the solubility of that compound in water increases. eg: Glucose ($C_6H_{12}O_6$) however, as the size of the side chain is increased, the influence of the polar functional group is less and water solubility decreases.

Branching of the alkyl chain lowers intra molecular forces between molecules and leads to increased solubility in water. eg: t-butyl alcohol is expected to be more soluble in water than n-butyl alcohol.

Compounds soluble in water are usually in all aqueous solvents and reagents. For eg: acetic acid is soluble in water, dilute HCl and dilute NaOH. But if a compound is only slightly soluble in water, it may be more soluble in another aqueous solvent. eg: a carboxylic acid such as butanoic acid may be only slightly soluble in water but very soluble in dilute bases. Acidic compounds which are insoluble in water, eg: phenol and benzoic acid, are insoluble in dilute HCl and are soluble in dilute NaOH solution forming sodium salts. Therefore solubility in NaOH is an indication of the acidity of these compounds. Acidification of the basic solution with dil. HCl will re-precipitate the acid.

$$RCOOH + NaOH \longrightarrow RCOO^-Na^+ \longrightarrow RCOOH$$

Aniline is a basic compound which is insoluble in water and forms a separate layer. It is insoluble in dil. NaOH and soluble in dil. HCl forming aniline hydrochloride. Basification of the acidic solution will release the basic compound and form an emulsion or separate layer.

$$Ph\text{-}NH_2 + HCl \longrightarrow Ph\text{-}NH3^+Cl^- \xrightarrow{NaOH} Ph\text{-}NH_2$$

Phenols are acidic enough to dissolve in dil. NaOH, but are not acidic enough to liberate CO_2 and dissolve in a solution of either dil. $NaHCO_3$ or Na_2CO_3. But nitrophenols are highly acidic and dissolve in both these reagents.

Acetone is a neutral organic compound which is soluble in water and is therefore soluble in dil. NaOH and dil. HCl. But neutral compounds such as ethyl benzoate and acetophenone are insoluble in water, dil. Bases and dilute acids.